

Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

AI in Customer Service: Efficiency and Satisfaction

Priya Singla (2210992097)

Priyank Mahajan (2210992099)

Priyanshi (2210992101)

Rudra Narayan Biswal (2210992191)

Supervised By

Dr. Shikha Tuteja, Associate Professor,
Computer Science & Engineering, Chitkara University, Rajpura

Department of Computer Science and Engineering,
Chitkara University, Punjab

Abstract and Overview



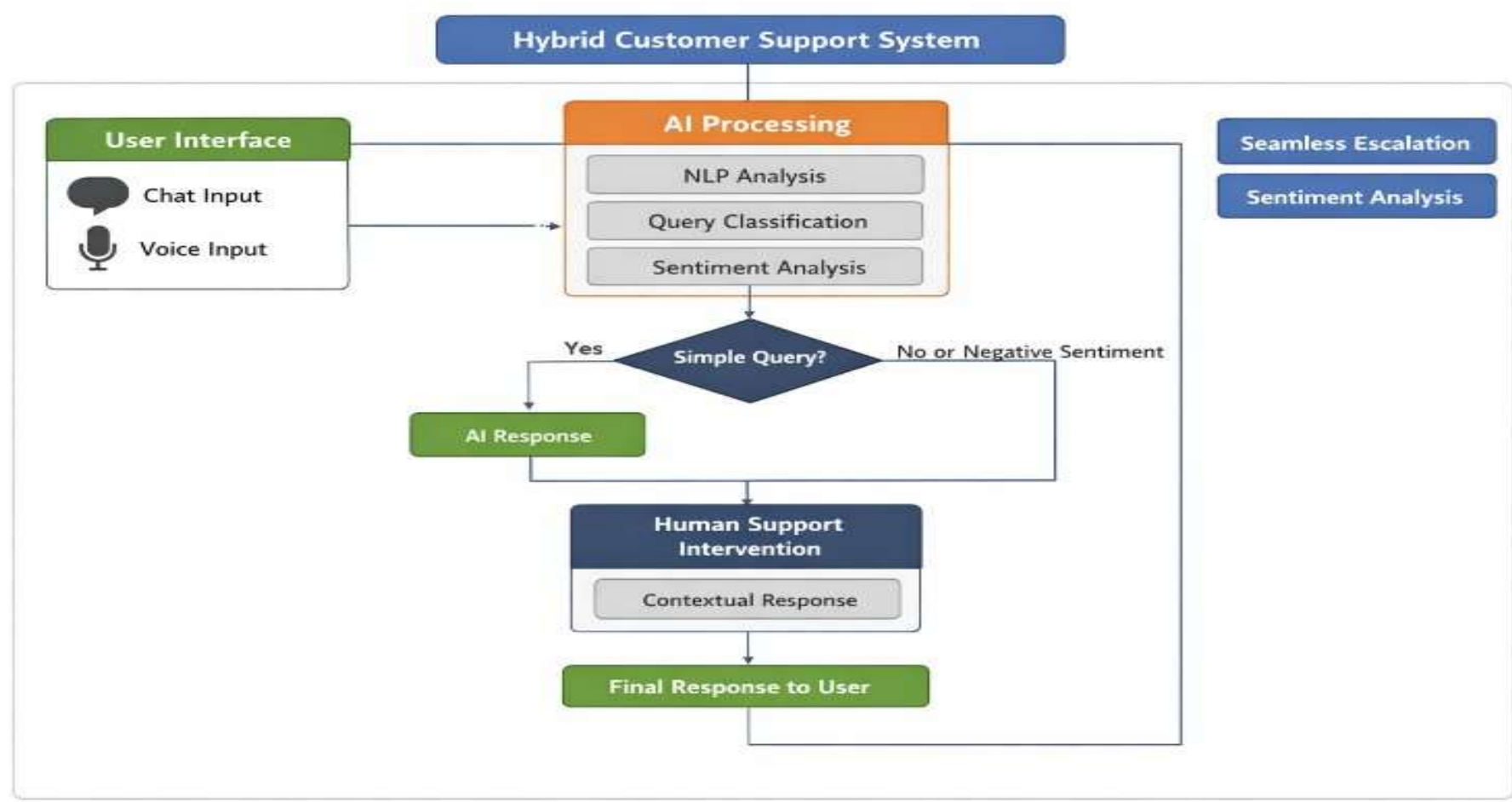
- Today's AI-powered customer service systems are reshaping how businesses interact with their customers. Handles high volumes cost-effectively with faster response and greater personalization.
- The system features **cross-lingual capability** to automatically translate user messages and reduces operational costs and error rates.
- Hybrid approach: AI automation combined with deliberate human intervention.
- Sentiment analysis routes complex queries to humans with seamless context transfer.
- Measured by response time, First Contact Resolution (FCR), and CSAT scores.
- AI alone is insufficient — the combination of efficiency and human judgment creates sustainable customer relationships.

- AI technologies have evolved from experimental tools to core business operations in customer service.
- Powered by advanced algorithms and natural language processing, these systems handle everything from routine enquiries to real-time multilingual support fundamentally changing what modern customer service looks like.
- Study examines efficiency gains: automation, cost reduction, and faster resolution.
- Also evaluates user-experience impact: personalization and customer satisfaction.
- Critically assesses strengths and weaknesses to build effective hybrid systems.
- Understanding where **AI excels** and where it falls short is the foundation for building systems that genuinely work.
- Key question: Where does AI excel, and where must humans intervene?

- **AI Boosts Efficiency:** Chatbots process large inquiry volumes without human intervention — faster and cheaper.
- Voice technology provides 24/7 assistance, eliminating queue wait times entirely.
- **Satisfaction Is Complex:** AI excels at immediate responses and recommendations.
- However, customers prefer humans for emotionally complex or ambiguous situations.
- **Hybrid Systems Win:** Blending AI automation with human assistance outperforms either approach alone.
- The research is clear that these systems outperform purely automated or purely human approaches on both efficiency and satisfaction metrics.
- Lack of transparency in AI decision-making is a significant ongoing concern.

- Three-tier architecture:
 1. User Interface Level (Chat & Voice).
 2. AI Processing Level (Intent recognition, query classification).
 3. Human Support Level.
- Step 1: User submits request via chat or voice interface.
- Step 2: AI applies Natural Language Processing (NLP) to understand user intent.
- Step 3: Query is classified as simple or complex by intent/sentiment analysis.
- Step 4: Simple queries receive AI auto-response immediately.
- Step 5: Complex queries are escalated to human agents with full conversation context.
- Step 6: Human agent provides personalized, empathetic answer to the customer.

Fig. 1. Hybrid AI-Based Customer Support System Architecture and Workflow



Results: Impact on Operational Efficiency



- Handle times decrease significantly with AI automation across all query types.
- Voice bots eliminate queue time for routine inquiries, improving customer wait experience.
- Substantial financial savings achieved by automating repetitive and mundane tasks.
- AI handles unexpected volume spikes without requiring crisis management protocols.
- AI enhances human staff capabilities — it does not replace them; service quality improves.
- Organizations report measurable ROI within months of hybrid system deployment.

Results: Influence on User Satisfaction



- AI Excels At: Order tracking, password resets, and simple product queries with instant answers.
- AI Struggles With: Emotionally complex issues, ambiguous statements, empathy, and regional accents.
- Critical Drop Point: Satisfaction plummets when customers hit an AI dead-end with no escalation path.
- Hybrid Systems Win: Combines the speed of AI with the empathy and judgment of human agents.
- Context transfer ensures users do not have to repeat themselves during escalation.
- Overall satisfaction scores highest in hybrid deployments versus pure AI or pure human models.

Comparative Analysis: AI vs Human vs Hybrid



- Response Time: AI is very fast; Human is moderate; Hybrid delivers fast resolution
- Availability: AI and Hybrid offer 24/7 coverage; Human agents are limited by working hours
- Cost Efficiency: AI is high; Human is low; Hybrid achieves moderate-to-high efficiency
- Complex Queries: AI is limited; Human and Hybrid both handle complex issues strongly
- Personalization: AI uses data-driven models; Humans use experience; Hybrid is highly effective
- Empathy: AI is low; Human is high; Hybrid delivers a balanced empathetic experience

Research Gaps & Future Scope



- Limited Multimodal Support: Current tools focus on text/speech; video and AR channels unexplored
- Static Personalization: Most systems rely on static rules; deep learning user models are underutilized
- Nuanced Emotion Detection: Existing sentiment analysis misses subtle cues like polite frustration
- Real-time Adaptive Learning: Most systems cannot learn from user input during an ongoing session
- AI Explainability: Customers need to understand why AI makes decisions
transparency builds trust
- SME Scalability: Affordable hybrid AI solutions for small/medium enterprises remain scarce

Thank You